Vedsai Vangapandu Computer Science Student | AI & Software Development

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Profile

Computer Science student at MacEwan University with hands-on experience in **Python, C#, SQL, REST APIs, AI integration, and full-stack development**. Experienced in **Gemini API integration, machine learning, Python automation, Agile/Scrum, SDLC, unit testing, and debugging** through software projects and research. Passionate about developing **AI agents, microservices, and responsible AI-powered solutions** through collaborative software development.

Education

Bachelor of Science - Computer Science Sep 2023 – Dec 2027
MacEwan University
Minors: Cybersecurity, Marketing
Relevant Coursework: Software Engineering, File & Database Management, Data Structures & Algorithms, Practical Programming Methodology, Human-Computer Interaction

Skills

- **Programming & Development:** Python, C#, .NET, JavaScript, Kotlin, Node.js, SQL, REST APIs, Object-Oriented Programming (OOP)
- **AI, Machine Learning & Automation:** Gemini API, Generative AI, NLP, TF-IDF, K-Means Clustering, Regression, Python Automation, Pandas, NumPy
- **Backend, Databases & Cloud:** Relational Database Design, CRUD Operations, Firebase Authentication, Cloud Firestore, Cloud Functions, Cloud Messaging
- **Software Engineering & Tools:** SDLC, Agile/Scrum, Unit & Integration Testing, Debugging, Defect Resolution, JUnit, AndroidX Testing, ArchUnit, Git/GitHub, Gradle, OpenPyXL, Power BI, Tableau, Plotly, Figma, Privacy-Aware Design, Technical Documentation

Professional Experience

Research Assistant – World Bank Economic & Governance Data Research Feb 2026 – Present
MacEwan University
- Develop **Python automation workflows** integrating the **World Bank Indicators REST API** to retrieve, process, and validate large-scale economic and governance datasets.
- Automated the transformation of **7,700+ raw records into 5,350 analysis-ready observations** across **161 economies and five years**, reducing repetitive manual processing.
- **Debug and optimize Python scripts**, resolve API integration and data-quality issues, and maintain reusable validation workflows using **Python, OpenPyXL, and Excel**.

Research Assistant – Product Recall & Consumer Safety Research Nov 2025 – Present
MacEwan University
- Develop **Python-based data processing and automation workflows** using **Pandas, OpenPyXL, and Excel** to analyze **9,793 product recall records across 82 variables**.
- Automated approximately **9,700 longitudinal calculations** and developed **four reusable analytical measures**, improving processing efficiency and reproducibility.
- **Test, debug, and validate Python scripts** to resolve data-quality issues and generate analytical reports and visualizations using **Power BI, Tableau, and Plotly**.

Research Assistant – Organizational Purpose & Nonprofit Performance Research Apr 2026 – Aug 2026
MacEwan University
- Developed **Python automation and data-processing workflows** using **Pandas, NumPy, OpenPyXL, and python-docx** to analyze **170+ academic studies** with approximately 40 variables per source.
- Applied **machine learning and NLP techniques**, including **TF-IDF, K-Means clustering, and regression**, while debugging scripts, validating outputs, and automating research reporting.

Research Assistant & Product Designer – SAAF HCI Research Sep 2025 – Jul 2026
MacEwan University
- Designed and refined **interactive UI/UX prototypes in Figma**, translating user requirements into application designs and evaluating **eight persuasive design features**.
- Conducted **usability evaluations with five HCI experts**, incorporating feedback on **privacy, trust, and usability** into design improvements; first-authored a research paper accepted for **SeGAH 2026**.

Student Tutor Jan 2026 – Present

kihêw waciston Indigenous Centre, MacEwan University

- Provide individualized support in **technical and quantitative coursework**, applying **analytical problem-solving, troubleshooting, and clear communication** to explain complex concepts.
- Collaborate with **students and university staff**, adapting explanations to individual needs while managing multiple priorities and providing constructive feedback.

Projects

Duck Duck Defer – AI-Integrated Deferred Exam Management Application (Award-Winning Software Engineering Project)

Technologies: Kotlin, Jetpack Compose, Node.js, Firebase, Gemini API, REST APIs, Git/GitHub, JUnit, AndroidX Testing, ArchUnit

- Collaborated in a **five-member Agile/Scrum team** throughout the **SDLC** to develop an award-winning, full-stack application using **Kotlin, Jetpack Compose, Node.js, and Firebase**.
- Developed **modular application components** for role-based dashboards, exam scheduling, bookings, messaging, notifications, and account management.
- Integrated the **Gemini API via REST APIs** and Firebase services, including **Authentication, Cloud Firestore, Cloud Functions, and Cloud Messaging**.
- Performed **unit testing, integration testing, debugging, and defect resolution** using **JUnit, AndroidX Testing, and ArchUnit**, supporting application reliability and collaborative development.

Movie Rental Database Management System

C#, .NET, SQL, Windows Forms, Relational Database Design

- Developed a **database-driven application using C#, .NET, and SQL**, implementing **CRUD operations**, rental processing, inventory management, and transaction tracking.
- Designed **relational database schemas, ER diagrams, and SQL queries** using joins, nested queries, and aggregations; integrated database operations with a **C#/.NET interface**.
- Performed **software testing, debugging, and defect resolution** to improve application logic, query accuracy, and data integrity.

IknowServiceAI – AI-Powered Service Request Agent

Technologies: Python, FastAPI, React, JavaScript, PostgreSQL, SQL, Gemini API, REST APIs, Docker, pytest, Git/GitHub

- Developed a **full-stack AI agent application** using **Python, FastAPI, React, and PostgreSQL**, integrating frontend interfaces with backend **REST API microservices**.
- Integrated the **Gemini API** to automate service request classification, knowledge retrieval, and intelligent recommendations, incorporating **AI governance, privacy safeguards, human-in-the-loop approvals, and audit logging**.
- Implemented **automated unit and integration testing with pytest**, debugged frontend/backend integrations, and containerized services using **Docker**.

Organizations

Vice President – MacEwan AI Alliance (MAIA) Sep 2026 – Present

MacEwan University

- Collaborate with the **executive leadership team** to support initiatives in **artificial intelligence, machine learning, and emerging technologies**, promoting technical learning, student engagement, and innovation.
- Contribute to **strategic planning, cross-functional collaboration, and knowledge sharing**, supporting an inclusive AI-focused student community.

Director of Events, MacEwan Data Analytics Club Oct 2025 – Present

MacEwan University

- Coordinate **event planning, technical workshops, speaker sessions, and project activities**, collaborating with club executives, students, and external participants to support successful event delivery.
- Apply **Agile-inspired coordination, problem-solving, and organizational skills** to manage schedules, technical setup, resources, and multiple priorities while addressing logistical challenges.
- Facilitate **cross-functional collaboration, stakeholder communication, and knowledge sharing**, supporting technical learning initiatives and continuous improvement within the student community.

Student Volunteer Jan 2026 – Present

MacEwan International Badminton Club

- Collaborate with **club members and organizers** to coordinate weekly activities, manage equipment and event logistics, and support an organized, inclusive student environment.
- Demonstrate **teamwork, adaptability, problem-solving, and communication skills** while supporting club operations, resolving logistical challenges, and engaging with students.